

Complete Experiment Log

A Living Task-Transfer Graph for Auto-Research — pilot, brain medical imaging

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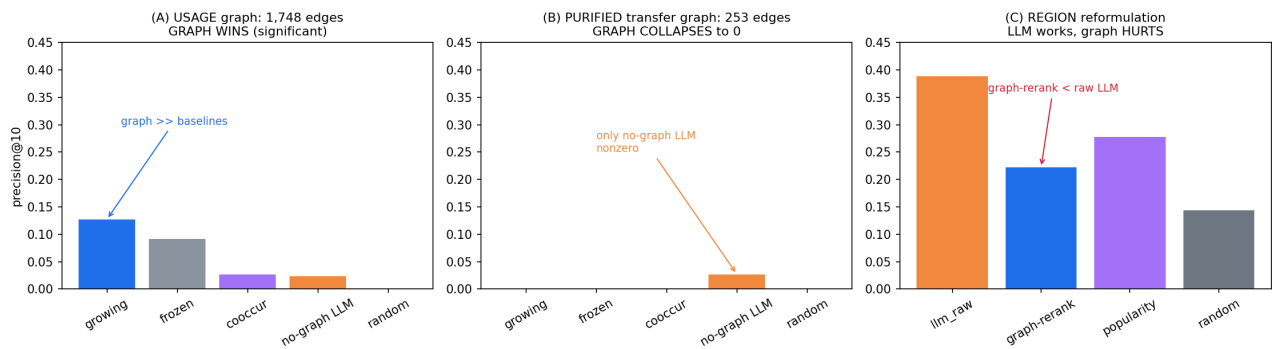


Figure. The whole arc in three regimes: (A) on the usage-heavy graph the graph navigation wins; (B) after purifying to genuine-transfer edges only, the graph collapses to 0; (C) under a region-level reformulation the LLM works but the graph does not help.

Overview of all experiments

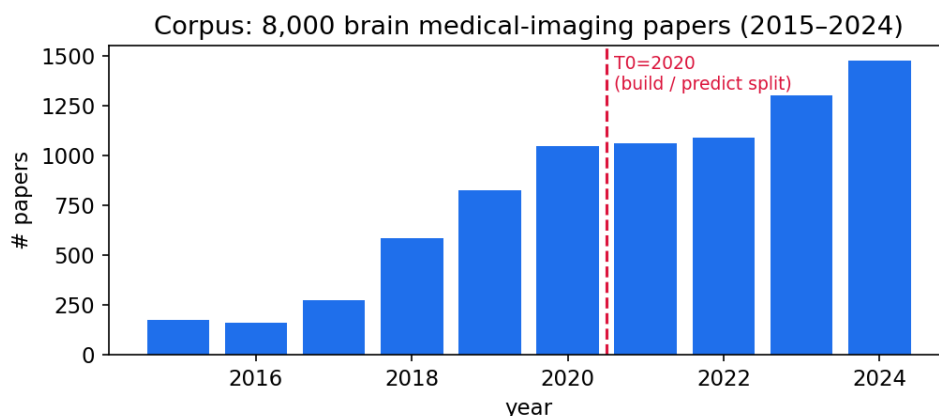
Exp	Question	Headline result
P0	Can we get a timestamped corpus?	8,000 CS brain-imaging papers, 2015-2024
P1	Does LLM extraction work?	yes - 40-paper sanity passes
P2	Build the graph	2,907 nodes / 1,748 edges; task->task only 8%
P4	Temporal compounding eval	exact-match metric -> all zeros (too strict)
P4b	Fix metric (semantic)	nonzero; compounding at small base
P5	k-sweep + 5 baselines	growing >= frozen > cooccur > no-graph > random
P6	Bootstrap CIs + significance	graph>baselines SIGNIFICANT; compounding NOT
P7	Full text vs abstract	prompt is the lever (8%->32%), not full text
P8	Edge quality audit	method->task 78% usage; only ~23% genuine transfer
P9	Transfer-prompt gate (30)	100% precision, 0% usage, but 83% empty
P10	Purify + re-eval (886)	1,748->253 edges; graph arms -> 0 (collapse)
P11	Region reformulation	tractable; LLM beats random; graph HURTS

P0 — Corpus acquisition

Goal. Obtain a timestamped brain medical-imaging corpus (corpus scope is a tunable parameter).

What we did. Semantic Scholar bulk search. The first query returned mostly clinical neuro-oncology (only 1,013/8,000 were Computer Science); re-queried with `fieldsOfStudy=Computer Science + method terms`.

	papers	venues found
result	8,000 (abstracts, 2015-2024)	arXiv, NeuroImage, ISBI, MICCAI, TMI, MedIA, SPIE



Takeaway. Corpus is a knob; with the CS filter we get the methods community, not clinical papers.

P1 — Extraction sanity (40 papers)

Goal. Validate the most uncertain component (LLM edge extraction) cheaply before scaling.

What we did. Run extraction on 40 papers (≤ 2020): task/method nodes + signed transfer edges + a deterministic post-check that the evidence span is a verbatim substring of the source.

task nodes	method nodes	transfer edges	dropped by post-check
75	131	26	0

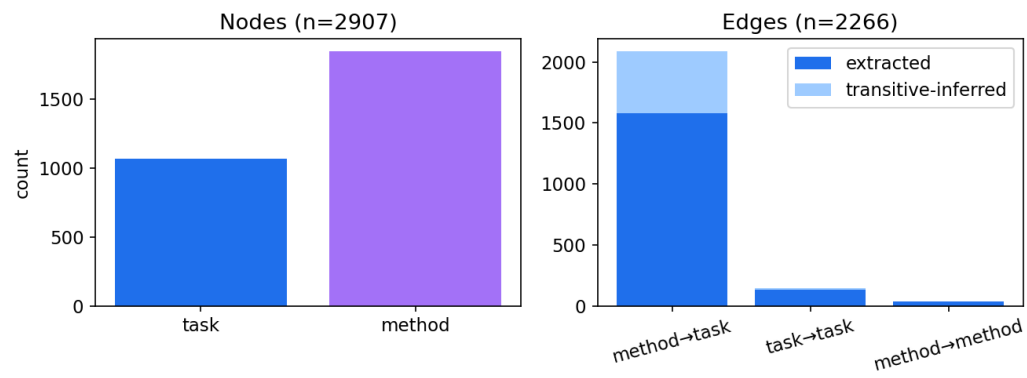
Takeaway. Extraction produces grounded, sensible edges (e.g. 'mixture-density network -> tumor growth estimation'). PASS.

P2 — Build the graph (886-paper subsample)

Goal. Construct the initial pilot graph.

What we did. Extract an 886-paper subsample (500 from ≤ 2020 + 100/yr 2021-24), entity-resolve near-duplicate nodes (MiniLM, $\cos \geq 0.86$), then transitive inference.

nodes	edges	method->task	task->task	method->method	+inferred
2,907	1,748	1,581	131 (8%)	36	+518



Takeaway. Graph builds cleanly, but task->task (the who-helps-whom edge) is only ~8% - first warning sign.

P4 — First temporal eval (exact match)

Goal. Test compounding: build graph $\leq T_0$, predict edges emerging in T_0+1 , score, then ingest.

What we did. Require an exact (source,target) node-id match among 2,907 nodes for a hit.

growing	frozen	random	diagnosis
0.000	0.000	0.000	only 5-13% of future edges are even predictable; 90% add a new node

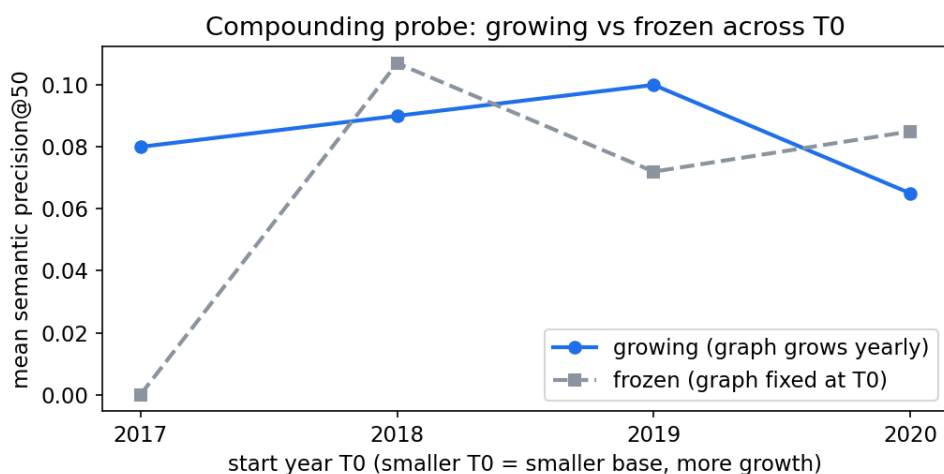
Takeaway. All zeros - but this is a metric problem (exact-id match is hopeless), not absence of signal.

P4b — Semantic-matching fix + T0 sweep

Goal. Make the metric measurable: a prediction hits if it is embedding-close to an emerged edge.

What we did. Semantic match ($\cos \geq 0.60$ on both endpoints). Sweep the start year T_0 from 2017 to 2020.

T0	2017	2018	2019	2020
growing	0.080	0.090	0.100	0.065
frozen	0.000	0.107	0.072	0.085



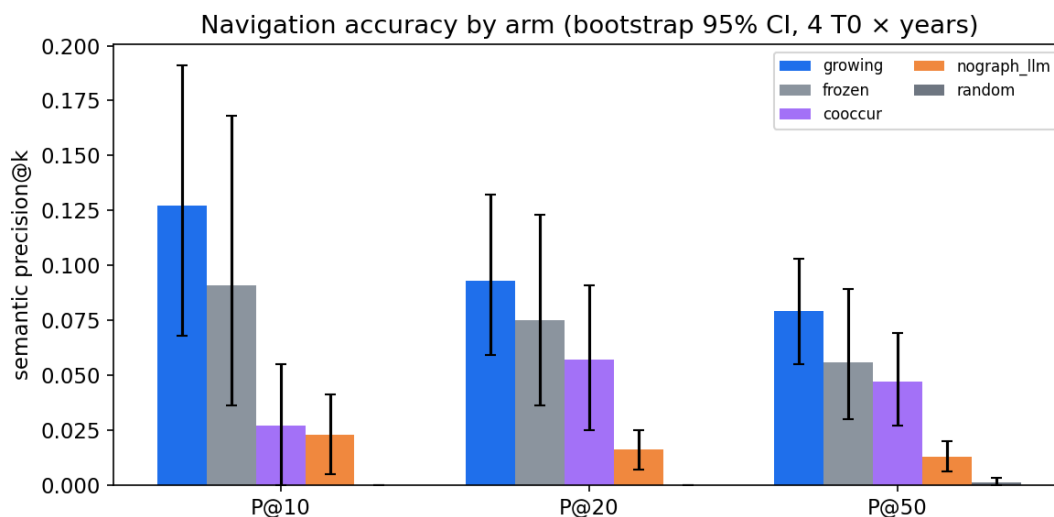
Takeaway. Nonzero now. Compounding (growing>frozen) shows at small base ($T_0=2017, 2019$), ties at large base.

P5 — Hardened eval: k-sweep + 5 arms

Goal. Compare all baselines across many k on the same temporal-forecast task.

What we did. Arms: growing / frozen / co-occurrence / no-graph-LLM / random; semantic precision@k for k in {5,10,20,50,100}.

P@10	growing	frozen	cooccur	no-graph LLM	random
mean	0.127	0.091	0.027	0.023	0.000



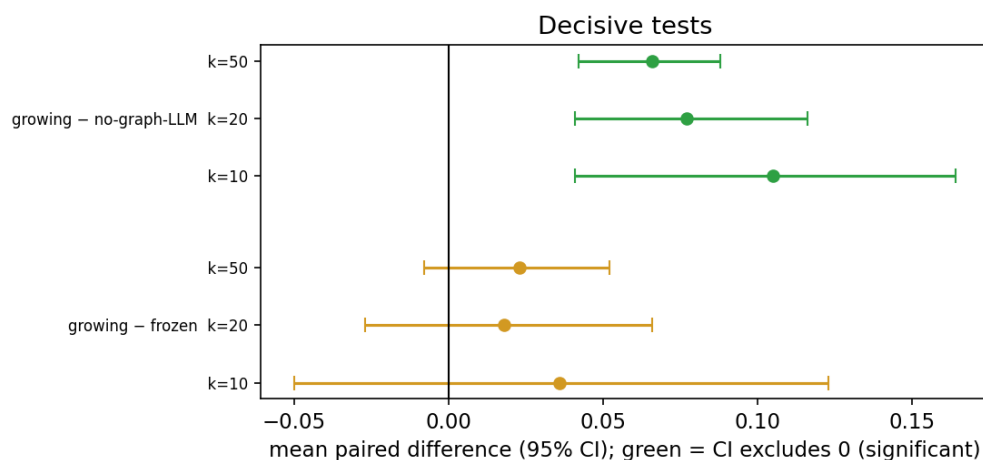
Takeaway. Consistent ordering: growing ≥ frozen > co-occurrence > no-graph-LLM > random. The graph beats everything; the LLM alone is near-random.

P6 — Bootstrap CIs + decisive paired tests

Goal. Decide whether the differences are real (CI excludes 0).

What we did. Pool 4 T0 settings x years; bootstrap 95% CI; paired tests for growing-frozen and growing-(no-graph-LLM).

paired test	k=10	verdict
growing - no-graph-LLM	+0.105 [+0.041, +0.164]	REAL (significant)
growing - frozen (compounding)	+0.036 [-0.050, +0.123]	NOT significant



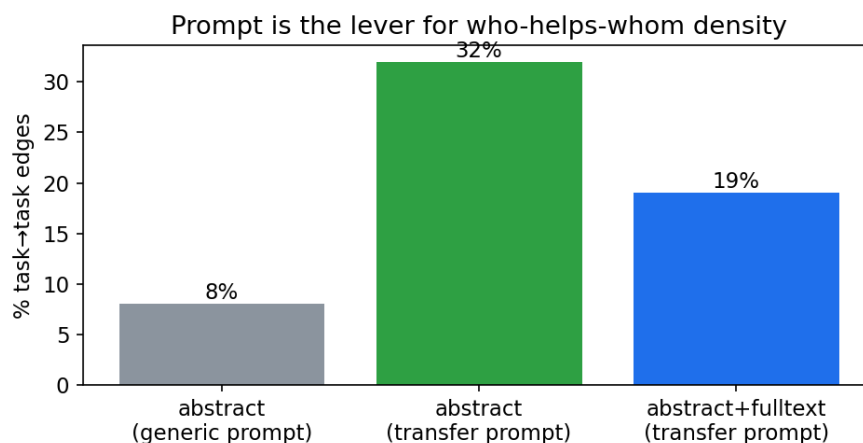
Takeaway. Graph > baselines is SIGNIFICANT. Compounding (growing>frozen) is directionally positive but NOT significant.

P7 — Full-text vs abstract A/B

Goal. Would reading full text thicken the thin task->task signal?

What we did. On 25 papers with PubMedCentral full text, extract from abstract-only vs abstract+full-text with a transfer-targeted prompt; count task->task share.

condition	task->task share	note
abstract (generic prompt)	8%	the original P2 graph
abstract (transfer prompt)	32%	prompt is the lever
abstract + full text	19%	full text dilutes (adds method->task)



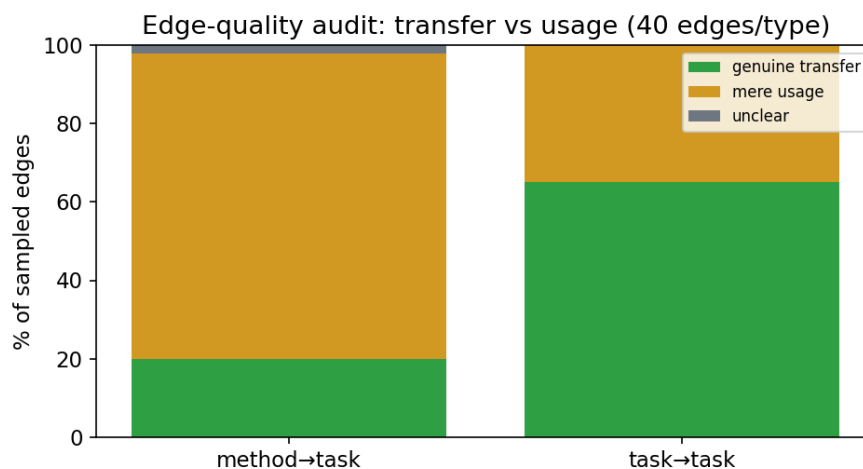
Takeaway. The extraction PROMPT, not full text, is the lever for who-helps-whom density.

P8 — Edge-quality audit (transfer vs usage)

Goal. Are the edges genuine transfer, or merely 'method used for task'?

What we did. An independent LLM judge classifies 40 sampled edges per type as transfer / usage / unclear.

edge type	genuine transfer	mere usage
method->task	20%	78%
task->task	65%	35%



Takeaway. The graph is largely a method-task USAGE table; genuine transfer is only ~23% of all edges.

P9 — Transfer-prompt validation gate (30 papers)

Goal. Before spending on a full re-extraction, prove a new prompt can purify usage out (precision-first gate).

What we did. Diagnostic 30 = 10 method-heavy + 10 task->task-heavy + 10 recent (2022-24). Two-stage transfer-vs-usage prompt; independent judge; thresholds: precision>=70%, usage<=25%, evidence>=95%.

true-transfer precision	usage leakage	evidence valid	empty rate
100% (12/12)	0%	100%	83%

Takeaway. Prompt cleanly separates transfer from usage (PASS on precision), but genuine transfer is very sparse (83% of papers yield none).

P10 — Purify + re-evaluate (full 886 papers)

Goal. Re-extract keeping only genuine transfer, then re-run the full evaluation on the clean graph (truth excludes inferred edges).

What we did. Re-extract 886 papers with the validated transfer prompt; entity-resolve; NO transitive inference; re-run all 5 arms with bootstrap CIs.

	edges	growing	frozen	cooccur	no-graph LLM	random
usage graph	1,748	0.127	0.091	0.027	0.023	0.000
transfer graph	253	0.000	0.000	0.000	0.027	0.000

Takeaway. Purifying to genuine transfer (1,748->253 edges) makes ALL graph arms collapse to 0; only the no-graph LLM stays nonzero. Diagnostic: navigation produces 0 candidates (1-2 'rich' nodes) and predictable-truth=0 (every emerged transfer edge introduces a NEW node). => The earlier win was carried by usage edges, not genuine transfer.

P11 — Region-level reformulation

Goal. Edge-completion is structurally impossible on a sparse, new-node-driven transfer graph. Reframe to predicting the semantic REGION of next-year transfer (new-node-tolerant).

What we did. Arms: LLM generates target areas (llm_raw); graph re-ranks the LLM list (llm_rerank); most-frequent past targets (popularity); random. Region-tolerant precision@k.

P@10	llm_raw	graph-rerank	popularity	random
mean	0.389	0.222	0.278	0.144
vs	rawLLM-random = +0.244 REAL		rerank-rawLLM = -0.167 (graph HURTS)	

Takeaway. The reformulation is tractable (0.3-0.5 vs 0.0 for edge-completion) and the LLM beats random - but the graph does NOT help (re-rank and popularity are below the raw LLM). Confounds: the LLM is leakage-inflated (recent cutoff), and the re-rank crudely replaced the LLM's ordering. A clean 'does the graph add value' test (LLM held constant, graph features ADDED not substituted) remains to be run.

Bottom line: what is established vs not

Established (real results):

- A complete, automated pipeline: fetch -> extract -> graph -> navigate -> temporal-replay evaluation.
- On the research-activity / usage graph, structure-driven navigation significantly beats no-graph-LLM, co-occurrence, and random (P6, with CIs).
- A history-as-label temporal-replay evaluation methodology (which the static-graph competitors cannot run).
- An honest edge-quality audit exposing the usage-vs-transfer distinction.

Not established:

- 'A genuine-transfer graph compounds (growing>frozen)' - after purification the transfer graph is too sparse and the graph collapses.
- 'The graph beats an LLM at discovering transfer' - not shown; the LLM is currently stronger.

Net. We built the system and obtained a real, significant result - but at the research-activity level. The moment we insist on genuine transfer, the signal is too sparse and the graph stops helping. The journey is an honest de-risking: a plausible compounding story, verified layer by layer down to where it actually holds.